

Aadish D Somayaji

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EDUCATION

Bachelor of Engineering in Computer Science

Atria Institute of Technology, Bangalore

CGPA: 9.02/10

2022 — Present

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL
- **Frontend:** React.js, Redux Toolkit, Tailwind CSS
- **Backend & APIs:** Node.js, Express.js, FastAPI, Flask, RESTful APIs, Google Gemini API
- **Databases:** PostgreSQL, Supabase, MongoDB, MySQL, Drizzle ORM
- **Cloud & DevOps:** AWS (ECS Fargate, EC2, SSM, ALB, Route 53), Docker, GitHub Actions, Nginx, CI/CD
- **Data & ML:** TensorFlow/Keras (LSTM), XGBoost, Scikit-learn, Pandas, NumPy, Whisper, Streamlit

EXPERIENCE

Full-Stack & AI Intern

Apr 2026 — Present

Reoxide, Bangalore (*Carbon Credit Marketplace Platform*)

- Built a Python/FastAPI ML microservice for carbon credit price recommendation, integrating an XGBoost model with an IRR-based financial ensemble; exposed results via a RESTful API consumed across 4 microservices.
- Engineered LLM-powered data extraction pipelines using Google Gemini API to parse PDD documents and extract structured ESG metrics; results stored and queried via PostgreSQL/Supabase.
- Designed and maintained relational schemas in PostgreSQL using Drizzle ORM; optimised SQL queries for subscription gating, marketplace listing lifecycle, and moderation approval workflows.
- Deployed services on AWS ECS Fargate with SSM Parameter Store for secrets management; set up GitHub Actions CI/CD pipelines for automated builds and deployments across all services.

Frontend Developer Intern

Feb 2026 — Mar 2026

Emovur, Bangalore

- Gained hands-on exposure to a production React codebase; studied component architecture, Git branching workflows, and team coding standards used in a live product environment.
- Assisted in API validation using Postman; observed and participated in cross-team data contract discussions between frontend and backend engineers.

PROJECTS

Blog Platform — Custom CMS with Admin Dashboard

Node.js, PostgreSQL, Docker, AWS EC2

- Designed a PostgreSQL relational database for posts, users, and sessions; implemented JWT authentication and bcrypt password hashing for secure access control.
- Deployed on AWS EC2 with Docker Compose, Nginx reverse proxy, and HTTPS via Let's Encrypt. Live: blog.aadishds.live

AudioScope — Real-Time Audio Processing Studio

React, Redux, Python, Flask, SciPy

- Built a full-stack DSP web app with a Flask REST API applying Butterworth IIR filters, Hann/Blackman window filters, Gaussian noise injection (SNR-controlled), and AM/FM modulation using NumPy and SciPy.
- Implemented in-browser microphone recording (MediaRecorder API) with server-side WebM→WAV transcoding via ffmpeg; visualised before/after waveforms using Plotly.js with Redux Toolkit for shared audio state. Live: golden-elf1f125c.netlify.app

Solar Power Forecast — End-to-End ML Forecasting Pipeline

Python, TensorFlow, FastAPI, GitHub Actions, Streamlit

- Trained an LSTM neural network on 92 days of ERA5 weather reanalysis data to predict hourly AC power generation up to 7 days ahead.
- Built an automated GitHub Actions pipeline to refresh forecasts hourly; deployed an interactive Streamlit dashboard for visualising predictions.

GDHelper — AI-Powered Speech Evaluation System

Python, Whisper, FastAPI, Docker

- Integrated OpenAI Whisper for audio transcription and embedding-based semantic similarity scoring for automated speech evaluation.
- Containerised the multi-service architecture with Docker for reproducible deployment; exposed results via a FastAPI interface.

MyNotes — Full-Stack Notes App

React Native, Express.js, MongoDB

- Built a full-stack mobile app with an Express.js backend and MongoDB for persistent storage; implemented CRUD operations with real-time UI updates.

PUBLICATIONS

- Melody Mixer: Novel Music Generation by Mixing Styles with Variational Autoencoders — *IEEE COMP-SIF 2025*
- AI-Driven Marine Species Identification and 3D Model Generation: A Cloud-Native Approach for Sustainable Aquatic Science — *Book Chapter, Sustainable Approaches in Aquatic Science (ISBN: 978-81-994425-4-2)*

CERTIFICATIONS

- NPTEL — Programming, Data Structures, and Algorithms Using Python
- NPTEL — Design and Analysis of Algorithms
- NPTEL — Introduction to Programming in C